

# Bin Cheng (程斌)

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## RESEARCH INTEREST

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My research interest lies in **quantum information science**, with a particular focus on

- Quantum algorithms in the NISQ/early fault-tolerant regime
- Quantum advantage
- Quantum computation and combinatorial structures
- Classical simulation of quantum systems
- Quantum error correction
- Quantum complexity theory

## EMPLOYMENT & EDUCATION

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| 2023–now  | <i>Postdoctoral Research Fellow</i><br>Centre for Quantum Technologies, National University of Singapore<br>Hosted by Patrick Rebentrost                                                                                                      |
| 2019–2024 | <i>Ph.D. in Computer Science</i><br>University of Technology Sydney (UTS)<br>Advisors: Zhengfeng Ji, Michael Bremner and Man-Hong Yung (external)<br>Ph.D. Thesis: <i>Classical Verification and Enhancement of Near-Term Quantum Devices</i> |
| 2014–2018 | <i>B.Sci. in Physics</i><br>Southern University of Science and Technology (SUSTech)<br>Advisor: Man-Hong Yung                                                                                                                                 |

## PUBLICATIONS & PREPRINTS

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1. Daniel J. Spencer, Bin Cheng, Dinh-Long Vu, Kishor Bharti, Alexey V. Gorshkov, Patrick Rebentrost  
**Pricing of graph-based quantum portfolios from first principles.** arXiv:2610.03279
2. Bin Cheng, Feng Pan  
**Exact Maximum Likelihood Decoding beyond Treewidth via Rank-Decomposition Dynamic Programming.** arXiv:2609.39556
3. Tongyin Lin, Bujiao Wu, Bin Cheng (**corresponding author**)  
**Asymptotically Good Quantum Codes with Addressable Transversal T Gates.** arXiv:2609.35129
4. Fei Meng\*, Bin Cheng\*, Jianan Li, Man-Hong Yung (\*: **equal contribution**, **corresponding author**)  
**Weak Permanent Anti-Concentration for Random Gaussian Matrices in Boson Sampling.** QCE 26, Toronto. arXiv:2607.22088
5. Jeong Yu Han, Bin Cheng, Dinh-Long Vu, Patrick Rebentrost. (**corresponding author**)  
**Quantum Advantage for Multi-option Portfolio Pricing and Valuation Adjustments.** Quantitative Finance (2026)
6. Zhenyu Chen\*, Bin Cheng\*, Minbo Gao, Xiaodie Lin, Ruiqi Zhang, Zhaohui Wei, and Zhengfeng Ji (\*: **equal contribution**)  
**Scalable Quantum Error Mitigation with Neighbor-Informed Learning.** arXiv:2512.12578
7. Bin Cheng, Ziyuan Wang, Ruixuan Deng, Jianxin Chen, and Zhengfeng Ji  
**Breaking the Treewidth Barrier in Quantum Circuit Simulation with Decision Diagrams.** QIP 2026, arXiv:2510.06775
8. Dinh-Long Vu, Bin Cheng, Patrick Rebentrost (**corresponding author**)  
**Low Depth Amplitude Estimation without Really Trying.** ACM Transactions on Quantum Computing (2025)
9. Ziyuan Wang\*, Bin Cheng\*, Longxiang Yuan, Zhengfeng Ji (\*: **equal contribution**)  
**FeynmanDD: Quantum Circuit Analysis with Classical Decision Diagrams.** In Computer Aided Verification (2025)
10. Michael J. Bremner, Bin Cheng, Zhengfeng Ji (**corresponding author**, **alphabetical order**)  
**Instantaneous Quantum Polynomial-time Sampling and Verifiable Quantum Advantage: Stabilizer Constructions and Classical Security.** PRX Quantum (2025)

11. Bin Cheng<sup>\*</sup>, Fei Meng<sup>\*</sup>, Zhi-Jiong Zhang, Man-Hong Yung (\*: [equal contribution](#))  
**Generalized Cross-Entropy Benchmarking for Random Circuits with Ergodicity.** Chip (2025)
12. Patrick Rebentrost, Alessandro Luongo, Bin Cheng, Samuel Bosch, Seth Lloyd (corresponding author)  
**Quantum Computational Finance for Martingale Asset Pricing in Incomplete Markets.** Scientific Reports (2024)
13. Xiao-Wei Li, Xiao-Ming Zhang, Bin Cheng, Man-Hong Yung  
**Reachability Deficit of Variational Grover Search.** Physical Review A (2024)
14. Bin Cheng, Xiu-Hao Deng, Xiu Gu, *et al.*  
**Noisy Intermediate-scale Quantum Computers.** Frontiers of Physics (2023)
15. Chong Ying<sup>\*</sup>, Bin Cheng<sup>\*</sup>, Youwei Zhao<sup>\*</sup>, He-Liang Huang, Yu-Ning Zhang, Ming Gong, Yulin Wu, Shiyu Wang, Futian Liang, Jin Lin, Yu Xu, Hui Deng, Hao Rong, Cheng-Zhi Peng, Man-Hong Yung, Xiaobo Zhu, Jian-Wei Pan (\*: [equal contribution](#))  
**Experimental Simulation of Larger Quantum Circuits with Fewer Superconducting Qubits.** Physical Review Letters (2023)
16. Fan Wang<sup>\*</sup>, Bin Cheng<sup>\*</sup>, Zi-Wei Cui, Man-Hong Yung (\*: [equal contribution](#))  
**Quantum Computing by Quantum Walk on Quantum Slide.** arXiv:2211.08659
17. Kai Sun, Zi-Jian Zhang, Fei Meng, Bin Cheng, Zhu Cao, Jin-Shi Xu, Man-Hong Yung, Chuan-Feng Li, Guang-Can Guo  
**Experimental Verification of Group Non-membership in Optical Circuits.** Photonics Research (2021)
18. Xi Chen<sup>\*</sup>, Bin Cheng<sup>\*</sup>, Zhaokai Li, Xinfang Nie, Nengkun Yu, Man-Hong Yung, Xinhua Peng (\*: [equal contribution](#))  
**Experimental Cryptographic Verification for Near-Term Quantum Cloud Computing.** Science Bulletin (2021)
19. Bujiao Wu<sup>\*</sup>, Bin Cheng<sup>\*</sup>, Fei Jia, Jialin Zhang, Man-Hong Yung, Xiaoming Sun (\*: [equal contribution](#))  
**Speedup in Classical Simulation of Gaussian Boson Sampling.** Science Bulletin (2020)
20. Man-Hong Yung, Bin Cheng (corresponding author)  
**Anti-Forging Quantum Data: Cryptographic Verification of Quantum Cloud Computing.** arXiv:2005.01510

21. Bin Cheng, Man-Hong Yung.

**Ubiquitous Complexity of Entanglement Spectra.** arXiv:1906.00324

## TALKS

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- **Weak Permanent Anti-Concentration for Random Gaussian Matrices in Boson Sampling**  
IEEE QCE 2026, in Toronto, Canada.
- **Breaking the Treewidth Barrier in Quantum Circuit Simulation with Decision Diagrams**  
29th Annual Quantum Information Processing Conference (QIP) 2026, in Riga, Latvia.
- **FeynmanDD: Quantum Circuit Analysis with Classical Decision Diagrams**  
Institute of Physics Singapore (IPS) meeting 2025, in Singapore.
- **Instantaneous Quantum Polynomial-time Sampling and Verifiable Quantum Advantage: Stabilizer Constructions and Classical Security**  
Joint CQT-Munich Center for Quantum Science and Technology (MCQST) Workshop 2025, in Singapore.  
Joint CS group meeting at CQT, 2024.  
CQT CS Seminar, 2023.
- **Experimental Cryptographic Verification for Near-Term Quantum Cloud Computing**  
Nagoya-SUSTech Quantum Information Workshop 2019, in Nagoya, Japan.  
Mini-Workshop on Quantum Verification 2019, in Shanghai, China.

## GRANTS

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- CQT Young Researcher Career Development Grant, Centre for Quantum Technologies, 2026–2028 (S\$15,000)  
Project title: *Fundamental Limits of Quantum Advantage in Boson Sampling Applications*

## PROFESSIONAL SERVICES

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- **Reviewer for conferences:** QIP (2026, 25, 24, 22, 20), TQC (2025, 24, 23), AQIS (2026, 25), QCE (2023, 21, 20), Asiacrypt 2020
- **Reviewer for journals:** Int. J. Quantum Inf.

## HONORS & AWARDS

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- PhD Scholarship from Sydney Quantum Academy, 2020-2023
- SUSTech Scholarship for Outstanding Undergraduate, 2016-2017
- University Start-up Scholarship by Shenzhen Municipal Government, 2014-2018
- SUSTech Scholarship for Outstanding Freshman, 2014

## TEACHING EXPERIENCE

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- TA, Mathematical Methods for Physicists
- TA, Quantum Mechanics I